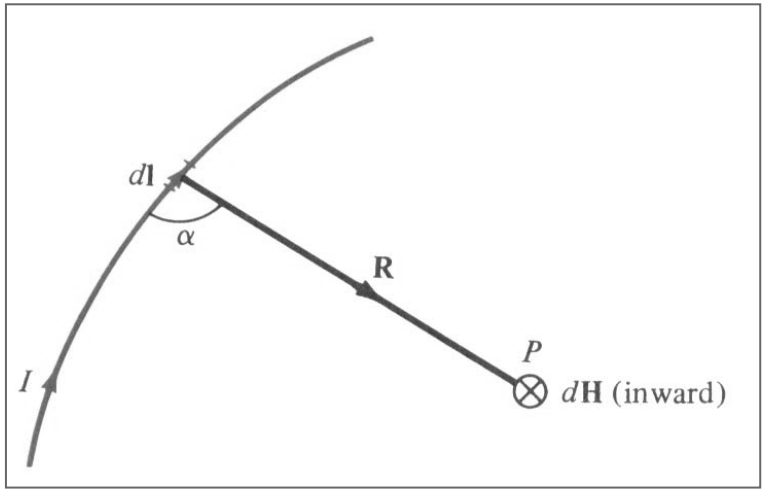
Static Magnetic Field: Biot-Savart's law

- Two major laws governs magnetostatic fields
  - Biot-Savart's law
  - Ampere's circuit law
- Biot-Savart's law
  - Biot-Savart's law states that the magnetic field intensity d**H** produced at a point P by a differential current element Id**l** is proportional to the product Id**l** and sine of the angle α between the element and the line joining P to the element and is inversely proportional to the square of the distance R between P and the element

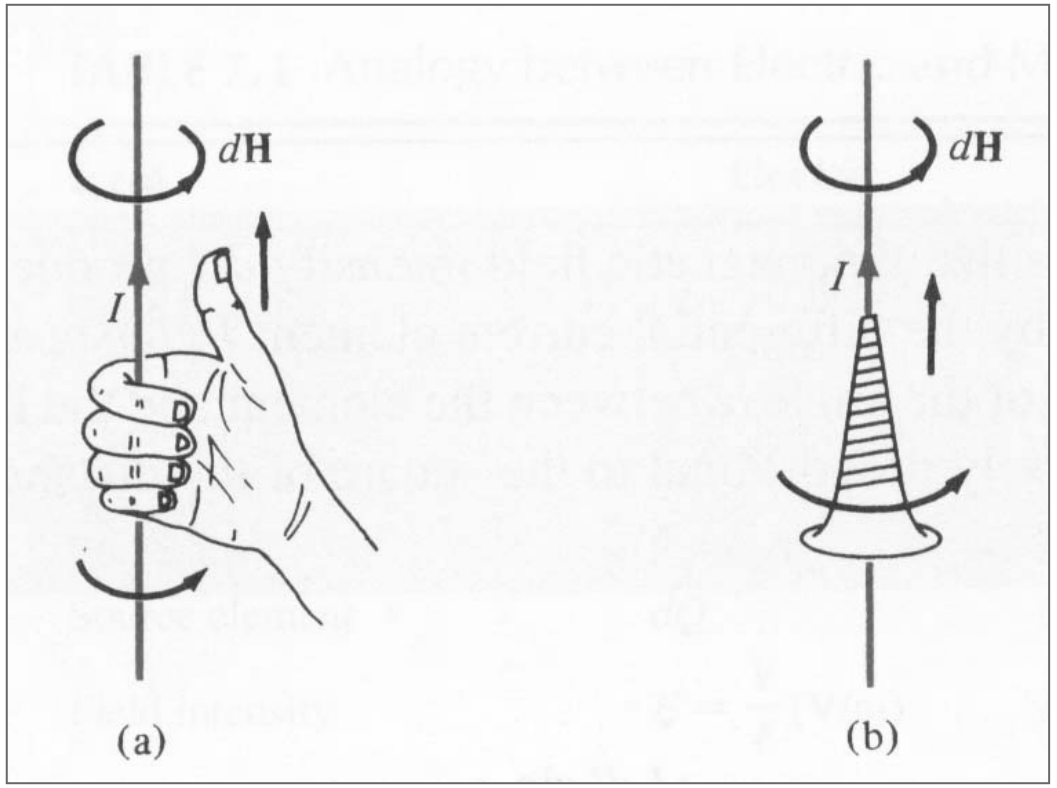
$$d\mathbf{H} = \frac{I\,d\mathbf{l} \times \mathbf{a}_R}{4\pi R^2} = \frac{I\,d\mathbf{l} \times \mathbf{R}}{4\pi R^3}$$

  - Where R=|**R**|, $\mathbf{a}_R$=**R**/R



Static Magnetic Field: Biot-Savart's law (cont'd.)

- The direction of d**H** can be determined by the right-hand rule
  - The right thumb points in the direction of the current, the right-hand fingers encircling the wires then indicate the direction of d**H**



Ampere's circuit law

- Ampere's circuit law states that the line integral of the tangential component of **H** around a closed path is the same as the net current I enclosed by the path

$$\oint_L \mathbf{H} \cdot d\mathbf{l} = I$$

- Applying Stoke's theorem

$$I = \oint_L \mathbf{H} \cdot d\mathbf{l} = \int_S (\nabla \times \mathbf{H}) \cdot d\mathbf{s}$$

Since $$I = \int_S \mathbf{J} \cdot d\mathbf{s}$$

$\therefore$ $$\nabla \times \mathbf{H} = \mathbf{J}$$

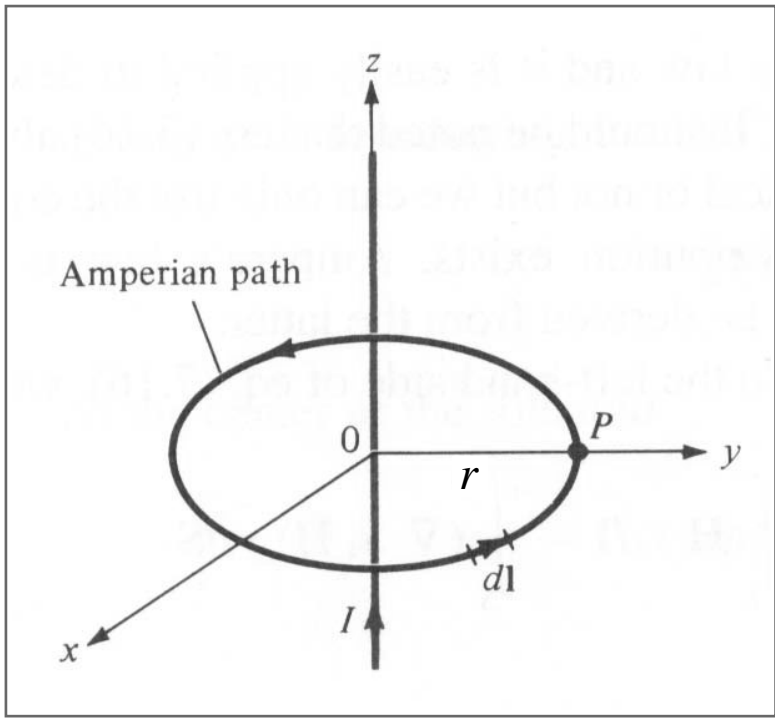
This is the third Maxwell's equation for static EM field

Example

Find the field at P due to a straight filamentary conductor

Solution:

- Choose a closed path passing through P
- Choose cylindrical coordinate
- H is a constant on the path



$$I = \oint_L H_\phi \mathbf{a}_\phi \cdot r d\phi \mathbf{a}_\phi = H_\phi r \oint_L d\phi = H_\phi 2\pi r$$

$$\mathbf{H} = \frac{I}{2\pi r}\mathbf{a}_\phi$$

Magnetic flux density

- In free space, the magnetic flux density **B** is related to the magnetic field intensity **H** by

$$\mathbf{B} = \mu_0 \mathbf{H}$$

  - Where μ_0 is the permeability of free space

$$\mu_0 = 4\pi \times 10^{-7} \ \text{H/m}$$

- The magnetic flux through a surface S is given by

$$\Phi = \int_S \mathbf{B} \cdot d\mathbf{s}$$

where the magnetic flux Φ is in webers (Wb) and the magnetic flux density is in webers/square meter (Wb/m^2) or tesla

Magnetic flux density (cont'd.)

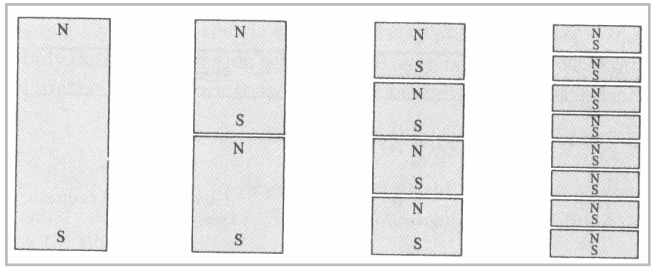
- **H is equivalent to E in static electric field; B is equivalent to D in electric field**
  - An electric charge can generate electric field which starts from a positive charge and ends at a negative charge

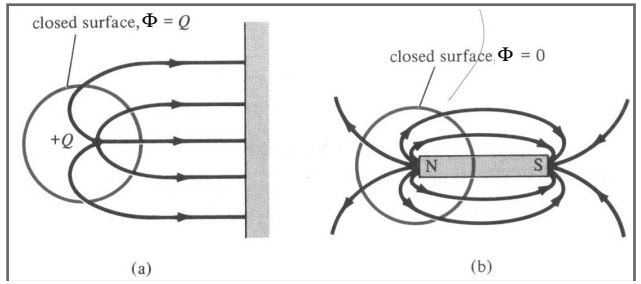
$$\oint_S \mathbf{D} \cdot d\mathbf{s} = Q$$

  - However an isolated magnetic charge does not exist (Magnetic poles always appear in pair)
    - Any closed surface must enclose the same "number" of North and South magnetic poles

$$\oint_S \mathbf{B} \cdot d\mathbf{s} = 0 \Rightarrow \oint_S \mathbf{B} \cdot d\mathbf{s} = \int_v \nabla \cdot \mathbf{B} dv = 0 \Rightarrow \nabla \cdot \mathbf{B} = 0$$

This is the fourth Maxwell's equation for static EM fields





Maxwell's equations for static EM fields

Differential (or Point) Form	Integral Form	Remarks
$\nabla \cdot \mathbf{D} = \rho_v$	$\oint_S \mathbf{D} \cdot d\mathbf{S} = \int_v \rho_v \, dv$	Gauss's law
$\nabla \cdot \mathbf{B} = 0$	$\oint_S \mathbf{B} \cdot d\mathbf{S} = 0$	Nonexistence of magnetic monopole
$\nabla \times \mathbf{E} = 0$	$\oint_L \mathbf{E} \cdot d\mathbf{l} = 0$	Conservativeness of electrostatic field
$\nabla \times \mathbf{H} = \mathbf{J}$	$\oint_L \mathbf{H} \cdot d\mathbf{l} = \int_S \mathbf{J} \cdot d\mathbf{S}$	Ampere's law

Vector magnetic potentials

- $\nabla \cdot \mathbf{B} = 0$ assures that **B** is solenoidal. Thus **B** can be expressed as the curl of another vector field, say **A**, such that

$$\mathbf{B} = \nabla \times \mathbf{A}$$

  - Vector **A** is the ***vector magnetic potential***
  - Its unit is Wb/m
- A vector is solely defined by its curl and divergence
  - We must define the divergence of **A**

$$\nabla \cdot \mathbf{A} = 0$$

  - Then

$$\left.\begin{aligned} \nabla \times \nabla \times \mathbf{A} &= \nabla \times \mathbf{B} = \mu_0 \nabla \times \mathbf{H} = \mu_0 \mathbf{J} \\ \nabla \times \nabla \times \mathbf{A} &= \nabla(\nabla \cdot \mathbf{A}) - \nabla^2 \mathbf{A} \end{aligned}\right\} \Rightarrow \nabla^2 \mathbf{A} = -\mu_0 \mathbf{J}$$

    - Vector Poisson's equation

Vector magnetic potentials (cont'd.)

- From $\nabla^2 \mathbf{A} = -\mu_0 \mathbf{J}$, it can be derived that

$$\mathbf{A} = \frac{\mu_0}{4\pi} \int_{V'} \frac{\mathbf{J}}{R} dv' \quad \text{(Wb/m)}$$

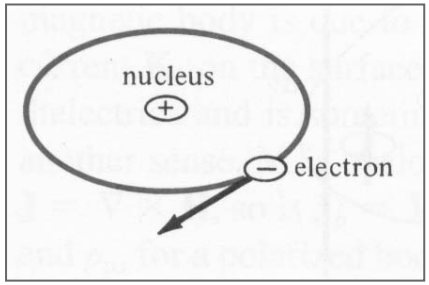
- **Physical significance** of **A** is that its line integral around any closed path is equal to the total magnetic flux passing through the area enclosed by the path

$$\Phi = \int_S \mathbf{B} \cdot d\mathrm{s} = \int_S \nabla \times \mathbf{A} \cdot d\mathrm{s} = \oint_L \mathbf{A} \cdot d\mathrm{l}$$

Magnetization in materials

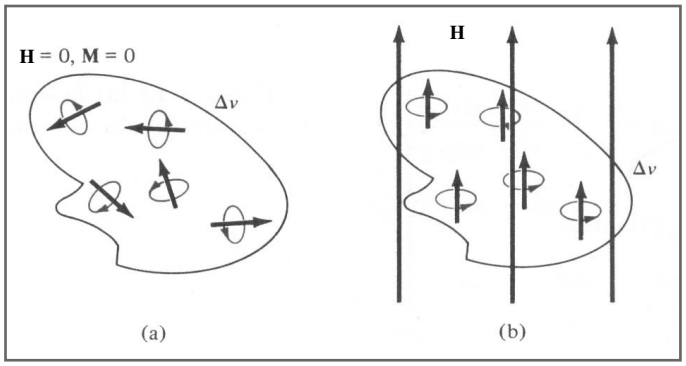
- All matter is composed of atoms
  - Electrons orbiting about a positive nucleus
  - The equivalent current loop has a magnetic dipole moment $\mathbf{m}_k$
  - An internal magnetic field is produced
- If there are n atoms per unit volume, a magnetization vector is defined as

$$\mathbf{M} = \lim_{\Delta v \to 0} \frac{\sum_{k=1}^{n\Delta v} \mathbf{m}_k}{\Delta v} \quad \text{(A/m)}$$



  - The **magnetization vector M** is the magnetic dipole moment per unit volume
  - Without an external **B** field, $\mathbf{m}_k$ is randomly oriented $\Rightarrow \mathbf{M} = 0$
  - When an external **B** field is applied, the magnetic moment of the electrons more or less align themselves with **B**, so $\mathbf{M} \neq 0$

Magnetization in materials (cont'd.)



- For linear and isotropic medium, **M** is directly proportional to the magnetic field intensity:

$$\mathbf{M} = \chi_m \mathbf{H}$$

  - χ_m is the **magnetic susceptibility**
  - The magnetic flux density corresponding to **M** is $\mathbf{B}_m = \mu_0 \mathbf{M}$
- The total magnetic flux density inside the medium is

$$\mathbf{B} = \mu_0 \mathbf{H} + \mu_0 \mathbf{M} = \mu_0 \mathbf{H} + \mu_0 \chi_m \mathbf{H} = \mu_0 (1 + \chi_m) \mathbf{H} = \mu_0 \mu_r \mathbf{H} = \mu \mathbf{H}$$

$$\mu_r = 1 + \chi_m = \frac{\mu}{\mu_0}$$ is the **relative permeability** of the medium